

**A Memory Layer for Actionable Interpretability.** SDM ESTIMATORS provide  $\hat{p}(y | \mathbf{x})$ , a robust estimate of the predictive *uncertainty*; interpretability-by-exemplar against a database of labeled examples (*introspection*); and local *updatability* via changes to  $\mathcal{D}_{tr}$  and  $\mathcal{D}_{ca}$ .

## Challenges

**Language models (LMs) are poorly calibrated** in high-probability regions in the presence of covariate and label shifts.

Existing classes of calibration methods do not target approximately conditional selection (cf. marginal coverage/calibration), which is needed for LM applications.

LM parameter non-identifiability necessitates interpretability-by-exemplar (i.e., instance-wise data valuation), including for calibration methods.

## Solution

We introduce **SIMILARITY-DISTANCE-MAGNITUDE (SDM) activation functions and estimators**, which are more robust and interpretable estimators of the predictive uncertainty than those based on the SOFTMAX function.

SDM activations **decompose signals of epistemic (reducible) uncertainty** in a manner that is interpretable and actionable.

This provides a principled, data-driven substrate for approaching **selective classification, calibration, and out-of-distribution detection** with language models.

## Task: Multi-class Classification

**Classify** the input  $\mathbf{x} \in \mathcal{X}$  into one of  $C$  discrete classes,  $y \in \mathcal{Y} = \{1, \dots, C\}$

**Data:** Labeled training set,  $\mathcal{D}_{tr} = \{(\mathbf{x}_n, y_n)\}_{n=1}^N$  and a corresponding held-out calibration set,  $\mathcal{D}_{ca}$ , drawn from the same distribution.

**Setting:** The parameters of the underlying LM are accessible, but held fixed. A final-layer adaptor has access to the frozen hidden states across all sequence positions.

**Seek:** Prediction,  $\hat{y}$ , *and* an estimate of  $p(y | \mathbf{x})$ , the predictive uncertainty of that classification decision.

**Calibration:** Class- and prediction-conditional accuracy at least  $\alpha$  (near 1)

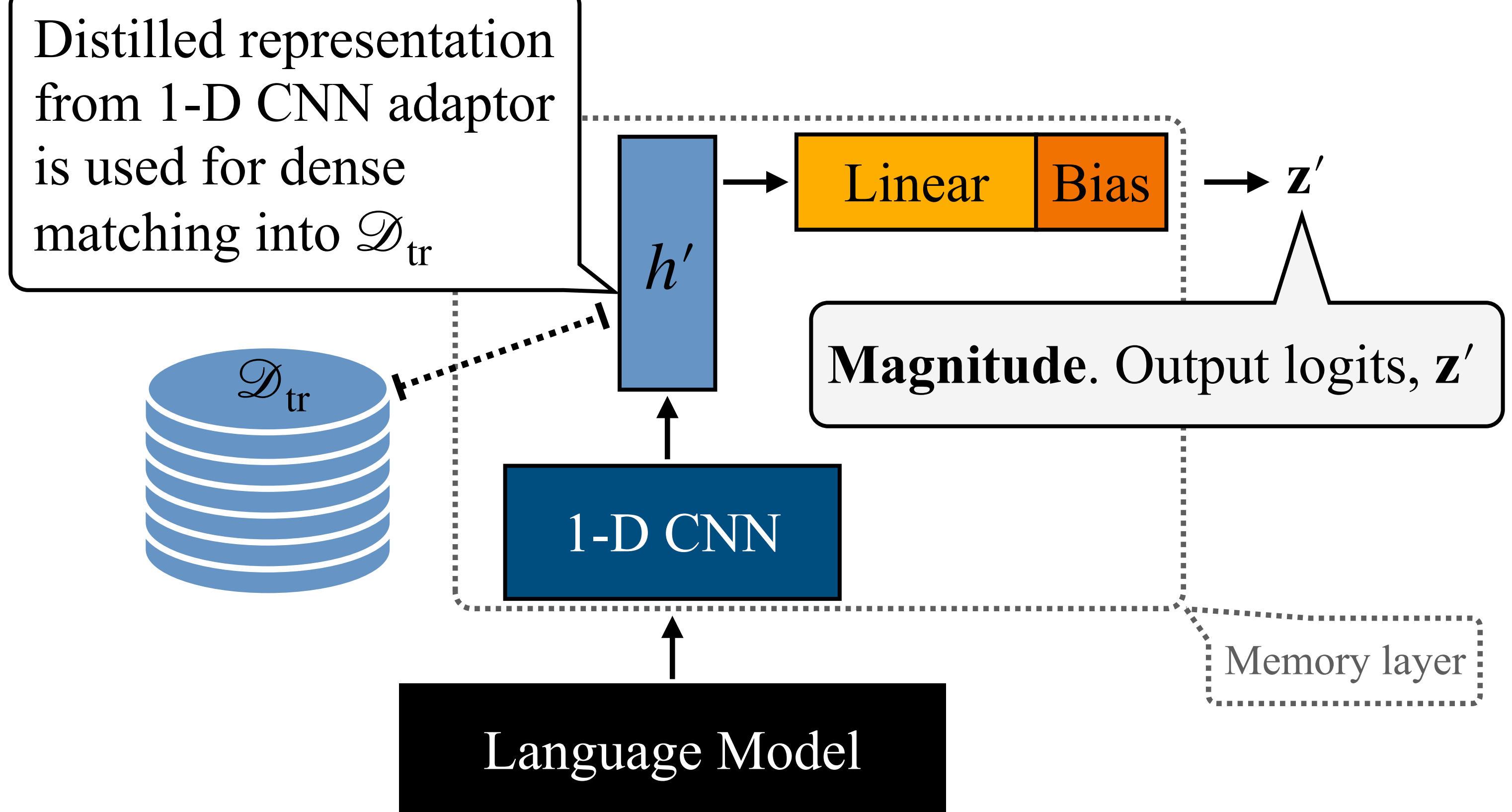
## Results

| $\alpha = 0.95$ |                    |                   | $y = 0$ |                                | $y = 1$ |                                | $\hat{y} = 0$ |                                | $\hat{y} = 1$ |                                | $y \in \{0,1\}$ |                                |
|-----------------|--------------------|-------------------|---------|--------------------------------|---------|--------------------------------|---------------|--------------------------------|---------------|--------------------------------|-----------------|--------------------------------|
| Dataset         | Model              | Estimator         | Acc.    | $\frac{n}{ \mathcal{D}_{te} }$ | Acc.    | $\frac{n}{ \mathcal{D}_{te} }$ | Acc.          | $\frac{n}{ \mathcal{D}_{te} }$ | Acc.          | $\frac{n}{ \mathcal{D}_{te} }$ | Acc.            | $\frac{n}{ \mathcal{D}_{te} }$ |
| SENTIMENT       | Phi3.5+ADAPTOR     | RAPS              | 0.98    | 0.45                           | 0.98    | 0.44                           | 0.98          | 0.45                           | 0.98          | 0.44                           | 0.98            | 0.89                           |
| SENTIMENTOOD    | Phi3.5+ADAPTOR     | RAPS              | 0.48    | 0.13                           | 0.82    | 0.18                           | 0.66          | 0.10                           | 0.68          | 0.22                           | 0.68            | 0.32                           |
| FACTCHECK       | Phi3.5+ADAPTOR     | RAPS              | 0.36    | 0.18                           | 0.98    | 0.35                           | 0.89          | 0.07                           | 0.74          | 0.46                           | 0.76            | 0.53                           |
| SENTIMENT       | Phi3.5+DISCVBLLMLP | VBLL              | 0.99    | 0.42                           | 1.00    | 0.40                           | 1.00          | 0.42                           | 0.99          | 0.40                           | 0.99            | 0.82                           |
| SENTIMENTOOD    | Phi3.5+DISCVBLLMLP | VBLL              | 0.89    | 0.03                           | 0.96    | 0.04                           | 0.93          | 0.02                           | 0.94          | 0.05                           | 0.94            | 0.07                           |
| FACTCHECK       | Phi3.5+DISCVBLLMLP | VBLL              | 0.35    | 0.07                           | 1       | 0.31                           | 1             | 0.02                           | 0.88          | 0.36                           | 0.88            | 0.38                           |
| SENTIMENT       | Phi3.5+SDM         | SDM <sub>HR</sub> | 1.00    | 0.37                           | 0.99    | 0.30                           | 0.99          | 0.38                           | 1.00          | 0.30                           | 0.99            | 0.68                           |
| SENTIMENTOOD    | Phi3.5+SDM         | SDM <sub>HR</sub> | 1       | <0.01                          | 1       | <0.01                          | 1             | <0.01                          | 1             | <0.01                          | 1               | 0.01                           |
| FACTCHECK       | Phi3.5+SDM         | SDM <sub>HR</sub> | R       | 0                              | 1       | 0.12                           | R             | 0                              | 1             | 0.12                           | 1               | 0.12                           |

Non-SDM estimators are poorly calibrated in the presence of covariate shifts and out-of-distribution data. *The standard SOFTMAX over logits is, in effect, a highly lossy compression of the epistemic uncertainty.*

**SDM estimators exhibit substantively improved robustness, while remaining informative over in-distribution data.**

## LMs as hidden instance-based metric learners



**SDM ACTIVATION.**  
Replaces SOFTMAX

$$\text{SDM}(\mathbf{z}')_i = \frac{\text{Similarity}^{\text{Distance} \cdot \text{Magnitude}_i}}{\sum_{c=1}^C \text{Similarity}^{\text{Distance} \cdot \text{Magnitude}_c}} = \frac{(2+q)^{d \cdot z'_i}}{\sum_{c=1}^C (2+q)^{d \cdot z'_c}}, 1 \leq i \leq C$$

**Distance quantile,  $d$ .**  
The DKW inequality is used to account for the effective sample size

$$d = \min \left( 1 - \text{eCDF}_{ca}^{y=0}(d_{\text{Nearest}}), 1 - \text{eCDF}_{ca}^{y=1}(d_{\text{Nearest}}) \right)$$

Test point: True label,  $y$ , unknown

Test:  $h'$

$L^2$  distance to nearest training instance

$d_{\text{nearest}}$

Comparisons to  $\mathcal{D}_{tr}$

$[\hat{y} = \hat{y}_{(1)}^{\text{tr}}] \wedge [\hat{y}_{(1)}^{\text{tr}} = y_{(1)}^{\text{tr}}]$

Train: 1st match

+

$[\hat{y} = \hat{y}_{(2)}^{\text{tr}}] \wedge [\hat{y}_{(2)}^{\text{tr}} = y_{(2)}^{\text{tr}}]$

Train: 2nd match

$[\hat{y} \neq \hat{y}_{(3)}^{\text{tr}}] \vee [\hat{y}_{(3)}^{\text{tr}} \neq y_{(3)}^{\text{tr}}]$

Train: 3rd match

$q = 2$

**Similarity,  $q$ .**

Count of consecutive (depth-wise) nearest matches in training that are correctly predicted *and* match  $\hat{y}$  of the test instance

$$\text{SDM}_{\text{HR}} := \begin{cases} \hat{y} & \text{if } q' \geq q'_{\min} \wedge \text{sdm}(\mathbf{z}')_{\hat{y}} \geq \psi_{\hat{y}} \\ \perp & \text{otherwise} \end{cases}$$

**SDM ESTIMATOR.**  
Data-driven partitioning of the calibration set to estimate a region with class- and prediction-conditional accuracy at least  $\alpha$ , a value near 1

$\text{SDM}_{\text{HR}}^{\text{lower}}$  models the effective sample size (see Appendix A.5)

